

Step 1 — Create Database

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with 'bookflow_db' expanded, showing 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The central pane shows the 'SQL File 5' editor with the following SQL script:

```
1 CREATE DATABASE bookflow_db;
2 USE bookflow_db;
```

The right pane displays the 'SQL Additions' window with the message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	09:56:14	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'; database exists	0.000 sec
2	09:56:28	USE bookflow_db	0 row(s) affected	0.000 sec

The bottom left pane shows the 'Table: books' with the following columns:

Column	Type
book_id	int AI PK
title	varchar(2)
isbn	varchar(1)
published_year	int

Step 2 — Create Books Table (DDL + Constraints)

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with 'bookflow_db' expanded, showing 'Tables', 'Views', 'Stored Procedures', 'Functions', and 'sys'. The central pane shows the 'SQL File 5' editor with the following SQL script:

```
3 INSERT INTO Books (book_id, title, isbn, published_year) VALUES
4 (1, 'The Great Gatsby', '9780743273565', 1925),
5 (2, 'To Kill a Mockingbird', '9780061120884', 1960),
6 (3, '1984', '9780451524935', 1949);
7 SELECT * FROM Books;
```

The right pane displays the 'SQL Additions' window with the message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom pane shows the 'Output' window with the following results:

#	Time	Action	Message	Duration / Fetch
1	10:00:35	Select output pane	3 row(s) returned	0.000 sec / 0.000 sec

The bottom left pane shows the 'Schema: bookflow_db' with the following tables:

Table	Columns
books	book_id, title, isbn, published_year

The bottom right pane shows the 'Result Grid' with the following data:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120884	1960
3	1984	9780451524935	1949

Step 3 & 4 — Insert Books and Display

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'bookflow_db' database with a 'books' table. The central pane shows the SQL query editor with the following code:

```
9 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
10 (1, 'The Great Gatsby', '9780743273565', 1925),
11 (2, 'To Kill a Mockingbird', '9780061120884', 1960),
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;
```

The 'Result Grid' shows the output of the SELECT query:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1983
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

The 'Output' pane shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 5, 6 & 7 — Members Table: Create, Insert, Display

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'bookflow_db' database with a 'members' table. The central pane shows the SQL query editor with the following code:

```
22 (101, 'John Smith', 'john.smith@email.com'),
23 (102, 'Emma Wilson', 'emma.wilson@email.com'),
24 (103, 'Michael Brown', 'michael.brown@email.com');
25 • SELECT * FROM Members;
26
```

The 'Result Grid' shows the output of the SELECT query:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

The 'Output' pane shows the execution of the queries:

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 8 — Create Loans Table (FOREIGN KEYS)

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'bookflow_db' database with its schema. The central pane shows the SQL script for creating the 'Loans' table. The script includes a primary key on 'loan_id' and two foreign keys: one for 'member_id' referencing 'Members(member_id)' and another for 'book_id' referencing 'Books(book_id)'. The right pane shows the 'SQL Additions' tab with a message about automatic context help. The bottom pane shows the 'Output' window with the execution results of the script.

```
22 (101, 'John Smith', 'john.smith@email.com'),
23 (102, 'Emma Wilson', 'emma.wilson@email.com'),
24 (103, 'Michael Brown', 'michael.brown@email.com');
25 SELECT * FROM Members;
26
27 CREATE TABLE Loans (
28     loan_id INT PRIMARY KEY,
29     member_id INT,
30     book_id INT,
31     loan_date DATE,
32     FOREIGN KEY (member_id) REFERENCES Members(member_id),
33     FOREIGN KEY (book_id) REFERENCES Books(book_id)
34 );
```

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id) REFERENCES Members(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id))	0 row(s) affected	0.062 sec

Step 9 & 10 — Insert 10 Loans and Display

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'bookflow_db' database with its schema. The central pane shows the SQL script for inserting 10 loans. The script includes a 'SELECT * FROM Loans;' statement. The right pane shows the 'SQL Additions' tab with a message about automatic context help. The bottom pane shows the 'Output' window with the execution results of the script. A 'Result Grid' is also visible, displaying the columns 'loan_id', 'member_id', 'book_id', and 'loan_date'.

```
38 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
39 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
40 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
41 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
42 SELECT * FROM Loans;
```

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0.1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id) REFERENCES Members(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id))	0 row(s) affected	0.062 sec
5	10:03:10	SELECT * FROM Loans LIMIT 0.1000	0 row(s) returned	0.000 sec / 0.000 sec

Step 11 — JOIN Query (Who Borrowed What)

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'bookflow_db' schema with tables 'loans' and 'books'. The central pane shows the following SQL query:

```
44
45 SELECT m.full_name AS Member_Name, b.title AS Book_Title
46 FROM Loans l
47 INNER JOIN Members m ON l.member_id = m.member_id
48 INNER JOIN Books b ON l.book_id = b.book_id
```

The 'Result Grid' shows the following columns: Member_Name, Book_Title. The 'Output' pane displays the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOR...	0 row(s) affected	0.062 sec
5	10:03:10	SELECT * FROM Loans LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
6	10:03:35	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.me...	0 row(s) returned	0.000 sec / 0.000 sec

Step 12 — GROUP BY Query (Books per Year)

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'bookflow_db' schema with tables 'loans' and 'books'. The central pane shows the following SQL query:

```
50
51 SELECT published_year, COUNT(book_id) AS Total_Books
52 FROM Books
53 GROUP BY published_year
54 ORDER BY published_year
```

The 'Result Grid' shows the following columns: published_year, Total_Books. The 'Output' pane displays the 'Action Output' table:

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOR...	0 row(s) affected	0.062 sec
5	10:03:10	SELECT * FROM Loans LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
6	10:03:35	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.me...	0 row(s) returned	0.000 sec / 0.000 sec
7	10:03:57	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER ...	3 row(s) returned	0.000 sec / 0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Shows the schema `bookflow_db` with tables `books` and `donation_history`.
- SQL File 5:** Contains the following SQL statements:

```
58 donation_id INT PRIMARY KEY,
59 book_id INT,
60 donor_name VARCHAR(100),
61 donation_date DATE,
62 FOREIGN KEY (book_id) REFERENCES Books(book_id)
63 );
64
65 START TRANSACTION;
66 INSERT INTO Books (book_id, title, isbn, published_year)
67 VALUES (4, 'Animal Farm', '9780451526342', 1945);
68 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
69 VALUES (1, 4, 'Raj Kumar', CURDATE());
70 COMMIT;
```
- Output:** A table showing the execution results of the statements.

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (book_id) REFERENCES Books(book_id))	0 row(s) affected	0.062 sec
5	10:03:10	SELECT * FROM Loans LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
6	10:03:35	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id	0 row(s) returned	0.000 sec / 0.000 sec
7	10:03:57	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
8	10:04:18	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Shows the schema `bookflow_db` with tables `books` and `donation_history`.
- SQL File 5:** Contains the following SQL statements:

```
68 VALUES (1, 4, 'Raj Kumar', CURDATE());
69 COMMIT;
70
71 CREATE INDEX idx_books_isbn ON Books(isbn);
72 SELECT * FROM Books WHERE isbn = '9780451524935';
```
- Result Grid:** Shows the results of the SELECT statement in the previous step.
- Output:** A table showing the execution results of the statements.

#	Time	Action	Message	Duration / Fetch
1	10:00:35	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	10:01:43	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
3	10:02:12	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
4	10:02:34	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (book_id) REFERENCES Books(book_id))	0 row(s) affected	0.062 sec
5	10:03:10	SELECT * FROM Loans LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
6	10:03:35	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id	0 row(s) returned	0.000 sec / 0.000 sec
7	10:03:57	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
8	10:04:18	COMMIT	0 row(s) affected	0.000 sec
9	10:04:49	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec